Hardware-In-the-Loop Robust Co-Design of an Underactuated System

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Problem

- Co-Design concurrently optimizes system morphology and control, e.g. to give stronger stability guarantees
- Problem: done almost entirely **in simulation**
  - Simulation might not capture the system's complexity
  - Changing morphology means that different system dynamics may dominate after optimization
  - Badly identified dynamics can corrupt the results

→ In practice, guarantees might not transfer to the physical system

Closing the Sim-to-Real Gap

- How can **real-world data** be used **in the loop** of the Co-Design process?
- How to minimize the **number of experiments** needed?
- How to **interpret** the sim-to-real **gap** algorithmically?

Parametrizing Robustness [1]

We consider a non-linear torque-controlled inverted pendulum with state $x = [\theta, \dot{\theta}]^T \in \mathcal{X} \subseteq \mathbb{R}^2$, input $u \in \mathcal{U} \subseteq \mathbb{R}$ and equilibrium $\tilde{x} = \left(\pi, 0\right)^T$. Linearizing the system and applying an LQR controller with gains Q, R yields a Lyapunov function $V(x) = x^T P x$ that guarantees stability of the linearized, unconstrained system.

For the full non-linear closed-loop system, stability only holds within a *region of attraction* (RoA) around $\tilde{x}$. Since the RoA is generally intractable to compute exactly, it is in practice often approximated as a sublevel set of V:

$$\mathcal{R}_a := \{x \in \mathcal{X} | x \to \tilde{x} \text{ for } t \to \infty\} \approx \{x | V(x) < \rho\}. \tag{1}$$

This estimate is valid for the largest ρ such that $V(x) > 0$ and $\dot{V}(x) < 0$ for all $x \in \mathcal{R}_a \setminus \{0\}$.

As the RoA gives a formal stability guarantee, its volume can serve as a proxy for robustness: a larger volume means a larger safe region of operation.

Co-Design Formulation

Parameters:

- Morphology: Mass m of the system
- Control: LQR-controller gains Q and R

Parametrization to avoid over-parametrization and ensure well-definedness of Q and R:

$$R = \text{const.}$$
$$Q = \begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix} = \begin{pmatrix} q_1^{\max} e^{-\bar{q}_1} & 0 \\ 0 & q_2^{\max} e^{-\bar{q}_2} \end{pmatrix} \tag{2}$$

Objective: Maximize the *volume* of the approximated RoA as a proxy for stability and robustness: With $V(x) = (x - \tilde{x})^T P (x - \tilde{x})$ and eigendecomposition $\frac{1}{\rho} P = W^T \Lambda W$, calculate

$$vol(\mathcal{R}_a) = |det(\sqrt{\Lambda} W)|^{-1} \frac{\pi}{2}. \tag{3}$$

Optimization Problem: The full optimization problem becomes

$$\max_{m, \bar{q}_1, \bar{q}_2} \; vol(\mathcal{R}_a) \tag{4a}$$
$$\text{subject to } \; m \in [m_{min}, m_{max}] \tag{4b}$$
$$\bar{q}_i \in \left[0, -log\left(\frac{q_i^{\min}}{q_i^{\max}}\right)\right] \tag{4c}$$

with $vol(\mathcal{R}_a)$ evaluated with respect to the system dynamics at parameters $m, \bar{q}_1, \bar{q}_2$.

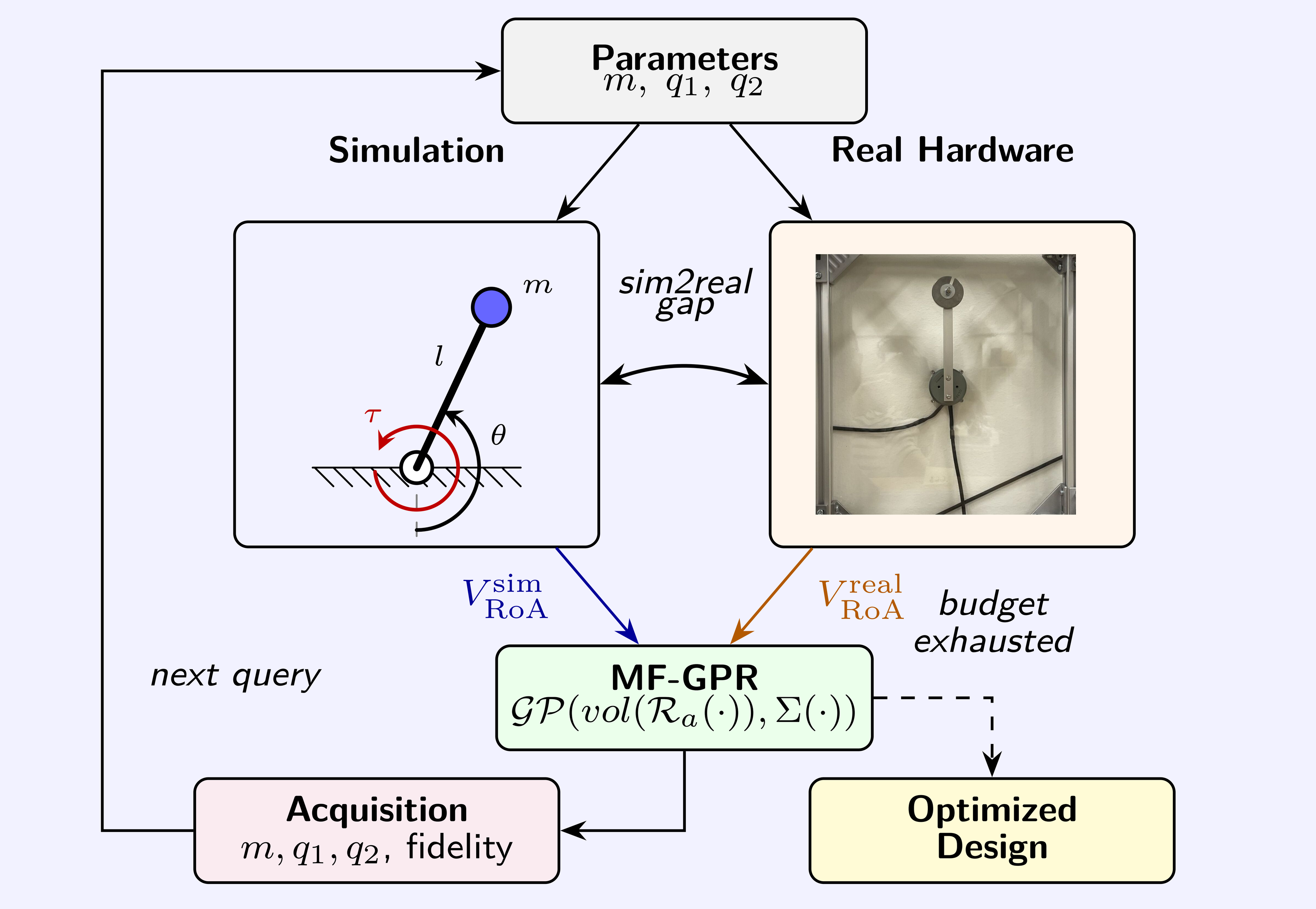
Optimization Algorithm:

- With a *gradient-free, sampling-based* optimization algorithm, the Co-Design formulation could run at *any fidelity level* (i.e. real system, different system models).
- We run it instead at *multiple fidelities concurrently*.

→ This allows to combine cheap simulations and more accurate experiments to balance cost versus accuracy

To this end, we adopt a Multi-Fidelity Gaussian Process Regression (MF-GPR) as described in [2]. The idea is to find a surrogate model for the map $f_i : (m, q_1, q_2) \mapsto vol(\mathcal{R}_a)$ at each fidelity i. The surrogate consists of multiple latent Gaussian Processes and a structure that describes the relationship between them. The step from value-estimation to optimization happens through a so-called acquisition function that selects the next test points that maximizes the information about the maximum of $f_i(m, q_1, q_2)$.

Framework Overview



Results

- Validation: torque-controlled simple pendulum ($m \in [70\,\text{g}, 150\,\text{g}]$, $l = 0.05\,\text{m}$, $u \in \pm 0.04\,\text{N m}$) using the CloudPendulum platform [3]
- 2 fidelities: "real" and "simulation"; simulation uses intentionally inaccurate model (no friction, damping, measurement noise)
- Initialized with 25 simulations and 3 experiments (cost 1 vs. 5) and runs for 52 iterations until the cost budget of 150 is exhausted (199 sims, 28 experiments / ca. $52\,\text{min}$ of experiment data)
- Result ($m^* = 70\,\text{g}$, $q_1^* = 7.3 \times 10^{-4}$, $q_2^* = 1.3 \times 10^{-6}$) aligns with physical intuition
- Validation against entirely simulation-based method (using CMA-ES [1]): $vol(\mathcal{R}_a^{MF-GPR}) = 47.43$ (ours) vs $vol(\mathcal{R}_a^{CMA-ES}) = 28.22$ (theirs)

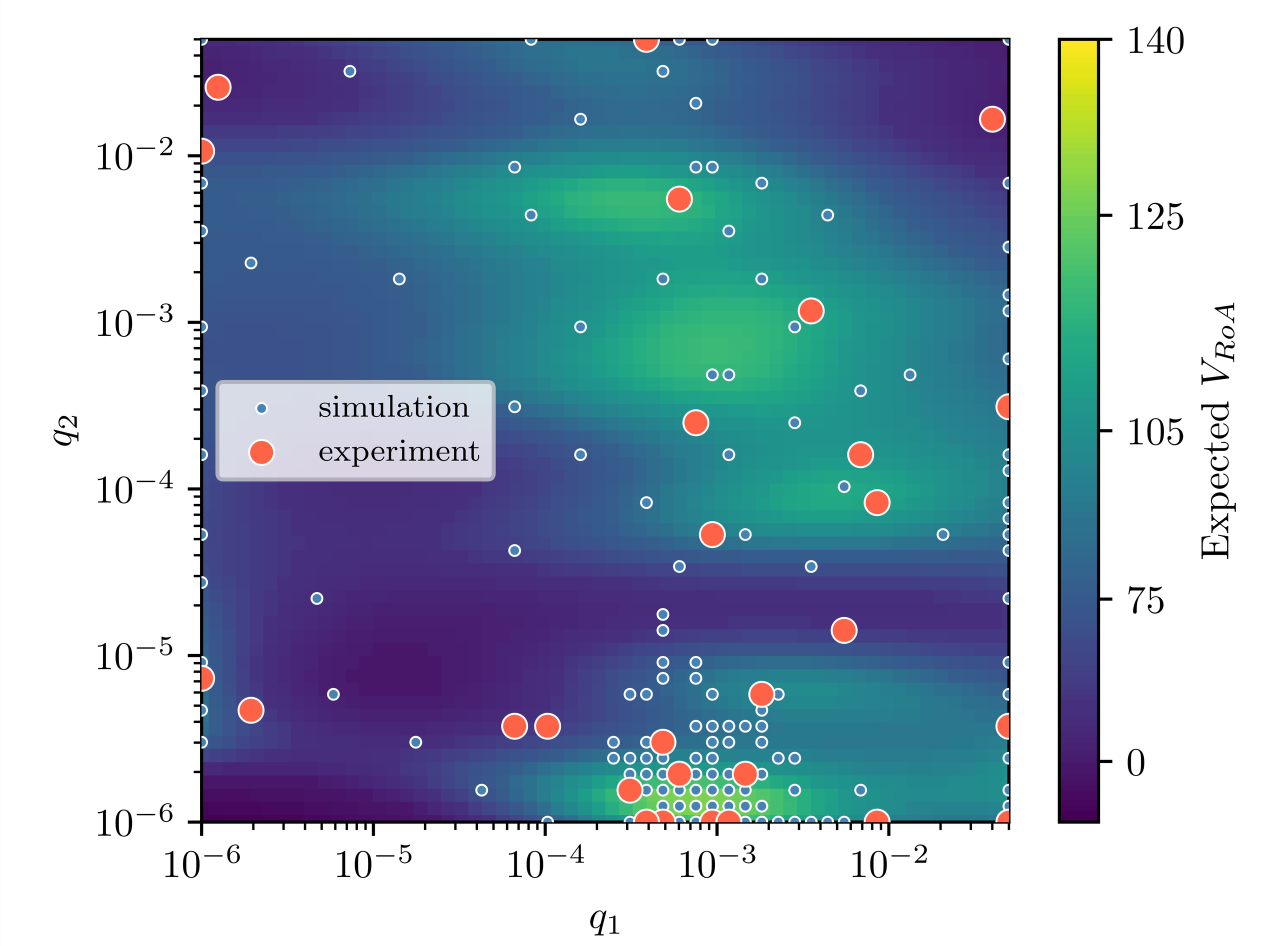


Figure 1: Estimated Volume of the RoA at "experiment" fidelity for q_1, q_2 with $m = 70\,\text{g}$.

Method	q_1	q_2
MF-GPR	7.3×10^{-4}	1.3×10^{-6}
CMA-ES	2.7×10^{-3}	3.2×10^{-4}
Baseline	1×10^{-4}	1×10^{-6}

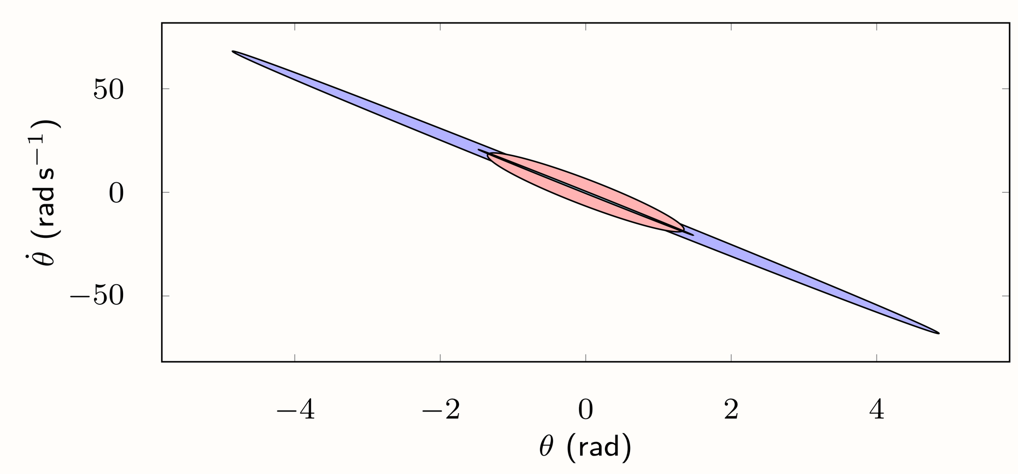


Figure 2: Resulting $\mathcal{R}_a$ as measured on the hardware, using $m = 70\,\text{g}$.

[1] Lasse Jenning Maywald et al. "Co-optimization of Acrobot Design and Controller for Increased Certifiable Stability". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022, pp. 2636–2641.

[2] Shion Takeno et al. "Multi-fidelity Bayesian optimization with max-value entropy search and its parallelization". In: *International Conference on Machine Learning*. PMLR. 2020, pp. 9334–9345.

[3] Shivesh Kumar. "Swinging pendulums on the cloud: Digitalization of simulation & experimental infrastructure for feedback-based active learning". In: *Chalmers Konferens om undervisning och lärande (KUL2025)*. 2025.

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